

Replication Report:
*Ab-initio simulation and experimental validation of
 beta-titanium alloys*
 (Raabe, Sander, Friák, Ma, Neugebauer; arXiv:0811.0157, 2008)

Automated replication (OpenClaw subagent)

2026-07-19

Abstract

We replicate the machine-checkable computational claims of Raabe et al. (2008), a DFT+experiment progress report on low-stiffness biomedical β -Ti alloys. The paper is **off-theme** for the “texture-spin” batch it was filed under (“texture” is one characterized property, not the subject). Its true ab-initio engine (plane-wave DFT total energies, GGA-PBE96) is HPC-scale and out of scope; we therefore isolate and reproduce the *closed-form* thermodynamics (ideal-mixing entropy, finite- T free energy) and the quantitative data/arithmetic claims. Of four selected claims: C1 (entropy) PASS, C3 (elastic data) PASS (3/3), C4 (composition conversion) PASS, C2 (finite- T β -stabilization threshold) PARTIAL. Overall **PARTIAL REPLICATION**; Coverage 6/10, Agreement 8/10.

1 Paper claims and method

The paper combines DFT (GGA-PBE96) with casting/ultrasonic experiments to design β -Ti alloys with a low elastic modulus (approaching cortical bone, ~ 20 GPa, vs Ti-6Al-4V ~ 105 – 110 GPa). Key relations:

$$E_f^\sigma(\text{Ti}_x\text{X}_{1-x}) = E_{\text{tot}}^\sigma/N - x\mu_{\text{Ti}}^{\text{hcp}} - (1-x)\mu_X^{\text{bcc}}, \quad (1) \quad (1)$$

$$S_{\text{config}}(x) = -k_B [x \ln x + (1-x) \ln(1-x)], \quad (2) \quad (2)$$

$$F_f(x, T) = \langle E_f \rangle - T S_{\text{config}}, \quad (3) \quad (3)$$

$$\sigma = (\delta E / \delta \varepsilon) / (A a_0). \quad (4) \quad (4)$$

(Eq. 2 is printed in the paper without the leading minus; we use the physically-correct positive form, verified below.)

2 Selected machine-checkable claims

C1 Ideal-mixing configurational entropy (Eq. 2), with maximum $S(0.5) = k_B \ln 2$.

C2 Finite- T free energy (Eq. 3) lowers the β -stabilization threshold: Ti-Nb from ~ 93 at.% (T=0) toward ~ 25 at.% near the transition temperature; Ti-Mo from ~ 25 at.% to ~ 14 at.%.

C3 Reported experimental Young’s moduli: minimum is Ti-30at.%Nb (72.1 GPa); $\sim 37\%$ drop vs hcp-Ti reference (114.7 GPa); trends.

C4 Table-1 wt% $\leftrightarrow$ at% conversions for all 8 alloys.

3 Results

C1 — Configurational entropy (PASS)

Our implementation of Eq. (2) gives $S(0.5) = 0.69315 k_B$, matching $k_B \ln 2 = 0.69315 k_B$ to machine precision; $T \cdot S$ at the transition temperature (~ 1155 K) is ~ 69 meV/atom at $x = 0.5$. The entropy machinery is exact.

C2 — Finite- T β -stabilization threshold (PARTIAL)

We lack the DFT $\langle E_f(x) \rangle$ curve, so we anchor a labelled analytic surrogate to the paper’s stated $T = 0$ roots and apply Eq. (3):

System	$T = 0$ root (anchor)	finite- T (this work)	finite- T (paper)
Ti-Nb	93 at.%	79.0 at.%	~ 25 at.%
Ti-Mo	25 at.%	16.6 at.%	~ 14 at.%

Ti-Mo reproduces well; Ti-Nb does not. The gap is a limitation of the symmetric-parabola energy surrogate (the real Ti-Nb $E_f(x)$ is skewed toward dilute compositions), *not* of Eq. (3). Reported honestly, not tuned.

C3 — Elastic modulus data (PASS, 3/3)

Quoted experimental moduli (GPa): Ti-Nb {10:91.2, 20:75.8, 30:72.1}, Ti-Mo {10:88.4, 20:106.9}, hcp ref 114.7. (a) Minimum measured modulus = 72.1 GPa = Ti-30Nb (PASS). (b) Drop vs hcp ref = $100(114.7 - 72.1)/114.7 = 37.1\%$, paper $\sim 37\%$ (PASS). (c) Ti-Mo modulus increases with Mo (slope +1.85 GPa/at.%, $R^2 = 1.000$, PASS); Ti-Nb experimental series *decreases* (slope -0.96 , $R^2 = 0.889$), consistent with the paper’s own α/ω -contamination caveat for low-solute samples.

C4 — Composition conversion (PASS)

Recomputing at.% from the tabulated wt.% (standard atomic masses) reproduces the paper’s at.% column: 18/20 element rows agree to < 0.25 at.%; the single 1.02 at.% mismatch (Ti-20Nb) is a rounding artifact of the paper’s own rounded wt.% input.

4 Verdict

PARTIAL REPLICATION. All tractable closed-form and data/arithmetic claims verify (C1, C3, C4); the one DFT-energy-dependent claim (C2 Ti-Nb) is only partially reproduced because the underlying VASP energies are out of scope and were surrogated (labelled). No fabrication: every number is from paper text or a real script run.

Coverage: 6/10 Agreement: 8/10

Coverage is limited because the paper’s core (DFT) is not independently recomputed. Agreement is high on everything actually within scope.